Fiscal Year:	FY 2009	**Task Last Updated:**	FY 02/01/2010
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Project Title:	DNA Damage Responses Induced By HZE Particles In Human Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	10/01/2004	**End Date:**	09/30/2009
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes at that time, per JSC/A. Chu-ARC (jvp 6/2009) NOTE: Received NCE to 9/30/2009 per PI (7/2008)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NNA05CM04G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In this project we propose to verify the hypothesis that high energy HZE particles would generate a different and/or unique radiation response in mammalian cells as compared to X-rays. The primary objective of this project is to verify the hypothesis that high energy HZE particles would generate a different and/or unique radiation response in mammalian cells as compared to X-rays. Further more, we will verify whether HZE radiation would induce a bystander effect. Towards this end we will utilize recent advances in our knowledge of molecular mechanisms of DNA damage sensing, signaling, and repair, and changes in global gene expression patterns. Our specific aims are: 1) To test the hypothesis that HZE particles may induce a different gene expression profile than that induced by low LET radiation; 2) To verify the hypothesis that DNA double-strand breaks induced by HZE particles are persistently unrejoined in normal human fibroblasts; 3) To test the hypothesis that HZE particles would induce a persistent and/or altered molecular and cellular response than that induced by X-rays in mammalian cells; 4) To test the hypothesis that high HZE may generate factors that induce a bystander effect or bystander-like effect. We plan to use DNA-damage responses as end points in combination with a novel track-detecting device to fulfill these objectives. The proposed research should reveal the fundamental differences in cellular responses between low LET X-ray and HZE exposures in humans. These studies would, in turn, provide biological markers for risk assessment and guidance in radiation protection for astronauts during space missions in the future. Specific Objectives A.1. To test the hypothesis that HZE particles may induce a different gene expression profile than that induced by low LET radiation. 1) To determine global gene expression patterns induced by HZE and X-rays at a dose of equal survival level. Metabolic pathways and expression markers specific to HZE responses in normal human fibroblasts will be identified. Especially, we will focus on pathways/gene groups related to cell proliferation, stress responses, DNA repair and protein kinases. 2) To determine whether gene expression profiles induced by HZE will vary as a function of energy of the HZE particles. 3) To determine whether transfer of media from HZE-irradiated cells to non-irradiated cells would induce specific gene expression changes in the recipient cells. A.2. To verify the hypothesis that DNA double-strand breaks induced by HZE particles are persistently unrejoined in normal human fibroblasts. 1) To determine the RBE of gamma-H2AX phosphorylation and focus formation by HZE and X-rays at equal survival doses in normal human fibroblasts. 2) To determine whether gamma-H2AX phosphorylation is significantly more persistent at the sites of HZE-induced DSBs as compared to X-rays. 3) To determine whether the induction and persistence of H2AX phosphorylation and focus formation will vary as a function of energy of the HZE particles. A.3. To test the hypothesis that HZE particles would induce a persistent and/or altered molecular and cellular response than that induced by X-rays in mammalian cells. 1) To determine whether key DNA DSB-sensing protein kinases, DNA-dependent protein kinase (DNA-PKcs) and ataxia telangiectasia mutated (ATM), are persistently phosphorylated and activated upon HZE exposure in human fibroblasts. 2) To determine whether the key DSB-signaling and damage-induced cell cycle check point proteins (the MRN complex, MDC1, SMC1, Chk1, Chk2, and p53) are persistently localized on DNA damage sites and/or phosphorylated upon HZE particle exposure in human fibroblasts. 3) To determine if the interaction between homologous recombination (HR) and nonhomologous end joining (NHEJ) proteins with DNA lesions induced by HZE is different from that observed with X-rays. A.4. To test the hypothesis that high HZE may generate factors that induce a bystander effect or bystander-like effect. 1) To determine the extent of DSB induction mediated by bystander and bystander-like effects in HZE-irradiated cells, as measured by H2AX phosphorylation and focus formation and localization of DSB sensing and signaling proteins. 2) To determine whether bystander effects and bystander-like effects are mediated by ROS or modulated through lipid peroxidation in HZE-particle irradiated cells. 3) To determine whether DSB repair capability (HR and/or NHEJ) plays a role in modulating HZE particles-induced bystander and bystander-like effects in mammalian cells.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research will provide biological markers for risk assessment and guidance in radiation protection for astronauts during space missions in the future.
Task Progress:	During the last year of this project we obtained the following results: 1. To study induction and repair of clustered DNA damages, we irradiated HT1080 cells stably expressing green-fluorescent protein tagged XRCC1 (EGFP-XRCC1) with Fe, Si and Oxygen particles and immunostained with 53BP1 (double strand breaks) and OGG1 (base damage). Using these as surrogate markers, we noticed differential induction of clustered DNA damages by different HZE particles. Moreover, the number of cells harboring the un-repaired clustered damages was higher in Fe irradiated cells, as compared to Si and O. 2. To study the differential induction of different types of DNA damages, including DSBs, SSB and base damages, along the ion tracks traversed by HZE particles, we have established stable cells lines expressing RFP-XRCC1 and YFP-53BP1 and confirmed their recruitment to the sites of DNA lesions generated by Fe, Si particles and gamma-rays. Using this approach, we directly monitored induction of single and DSBs in single living cells. 3. To study the genomic instability caused by charged particles, we irradiated HT1080 cells with Fe particles (1Gy, 1GeV/n) and examined the chromosomal aberrations 24 hours after irradiation. When we analyzed the metaphase spreads, we noticed different types of chromosomal abnormalities in Iron irradiated cells as compared with mock-irradiated cells. 4. To determine the cell cycle delay induced by charged particles, we irradiated exponentially growing HT1080 cells with Fe particles and collected samples before (pre-IR) and 0.5, 1, 2, 4, 8, 12, 24 and 48 hours after irradiation and checked for cell cycle delay by Flow cytometer. Our results showed that irradiation of cells with Fe ions causes extended period of G2/M arrest.
Bibliography Type:	Description: (Last Updated: 08/14/2018)

Abstracts for Journals and Proceedings	Asaithamby A, Oscar Gonzalez, Chen DJ. "Visualization of low-dose radiation induced DNA double strand-breaks in living cells." Presented at the Low Dose Program Investigators' Workshop, Washington D.C., January 21-23, 2008. Low Dose Program Investigators' Workshop, Washington D.C., January 21-23, 2008. , Jan-2008
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